

APARNA SINGH

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CAREER OBJECTIVE

An M.Sc Economics student specialising in Data Analytics. I am looking to start my career in the field of Data Analytics and Operations where i will be able to apply my knowledge and data driven problem solving abilities to real world problems. I can smoothly conduct analysis using Python, R programming and Sql to perform certain queries and bring out conclusions from it.

EXPERIENCE

May 2026-June 2026

Reserve Bank of India - Monetary Policy Department | Interest Rate Division

May 2026 to June 2026

- Analysed 160 monthly observations (2013–2026) of Indian households to examine shifts in Indian household savings across bank deposits, mutual funds, SIPs and small savings schemes.
- Used Excel and R Programming to clean, organise and structure datasets from RBI, AMFI and NSI to prepare a trend based analysis and visualisation.
- In conclusion, It was observed that mutual fund AUM increased from around ₹40 lakh crore in March 2023 to around ₹73 lakh crore in March 2026 (83% growth), while its share of the combined savings pool rose from 18% to 23%.

SKILLS

Technical : Python (Numpy, Pandas and Matplotlib) , R programming, Sql, Power bi, MS Excel, MS Word, Spss, Orange Mining tool.

Analytical : Data Analysis and Visualisation, Exploratory Data Analysis, Statistical modelling, Causal Inference

PROJECTS

Insider Threat Detection - Orange Mining tool and Python

August 2026

- Used and analysed 118,614 user activity records across 22 behavioural features applying kNN to identify the malicious user behaviour and the patterns of threat detection.
- Orange data mining tool and Python were used to build a kNN model and test its performance against Naive Bayes using 5 Fold Cross Validation. Achieved a 97% classification accuracy, with kNN, correctly detecting 3,734 malicious cases and outperforming the Naive Bayes model.

Automated Credit Risk Contagion Engine - Sql integrated python (SQLite)

November 2025

- Built a simulated credit risk stress testing engine covering 10,000 companies and 24,998 supply chain links to model how corporate defaults move across interconnected vendor networks.
- Stress testing showed a single-company default triggered 2,910 additional supplier failures, resulting in 2,911 total defaults and a 29.11% systemic contagion rate across the network.

Correlation Between Traffic Congestion and Neighborhood Air Quality - Python

October 2025

- Analysed approximately 500 hourly observations by combining Pune traffic volumes with PM2.5/NO₂ air-quality readings across three major intersections of Pune.
- The analysis indicated a positive relationship between traffic intensity and air pollution, with high-traffic periods associated with roughly 15–20% higher PM2.5 levels than low-traffic periods.
- Traffic-heavy locations showed consistently poorer air quality, suggesting that local traffic conditions contribute meaningfully to neighbourhood-level pollution disparities.

Analysed the will to pay for sustainable Goods - Rstudio

August 2026

- Conducted an experiment of 20 product choices across Pune which included necessity, normal and luxury goods to study people's preferences.
- Used Multinomial Logit and observed a clear gap between the stated preference and the actual will to pay. People refused to pay more for the Sustainable goods.

EDUCATION

M. Sc. Economics (Data Analytics)

Expected 2027

Symbiosis School of Economics

B. Sc. Economics and Statistics

2025

St. Joseph's University - 7.4 GPA

CERTIFICATIONS

Applied Text Mining in Python- University of Michigan

Social Media Data Analytics - University of Washington

Data Science in Python , IBM Generative AI- Coursera

Using R for Geostatistical Geospatial Modeling - Case Western Reserve University